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## Hydrodynamic Changes of Estuarine Islands in the Meghna River under Future Climate Scenarios

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Hydrodynamic processes strongly influence coastal and estuarine landscapes, especially in low-lying deltaic regions like the Meghna Estuary. Islands in the lower Meghna and at the Padma-Meghna confluence face increased risk of submergence due to sea-level rise and intensified precipitation under future climate scenarios. This study analyzes the long-term hydrodynamic changes in the Meghna Estuary using Delft3D simulations for 2000, 2024, and 2050, focusing on islands such as Nijhum Dwip, Moulovi Char, Domar Char, Char Kukri Mukri, Rajrajeshwar, Hatiya, and Manipura. The model covers the area from Baruriya Transit to the sea, integrating tidal and riverine dynamics. Future discharge for 2050 was generated from MIROC6 climate projections under SSP2-4.5 and SSP5-8.5 scenarios, bias-corrected and simulated via HEC-HMS. HEC-HMS was calibrated using 2022 data and validated with 2023 discharge records from Bhairab Bazar, while Delft3D was calibrated and validated using observed water level data from Daulatkhan over the same period. Results show rising tidal amplitudes and water levels, with high tides near Char Kukri Mukri increasing by 30 to 35 cm by 2050. Tidal inundation is expected to expand during monsoons, increasing flood risk in low-lying areas. Islands like Char Kukri Mukri and Hatiya are losing relative elevation, heightening their vulnerability to flooding and storm surges. Hydrodynamic projections indicate an average increase in water depth of 0.5 to 0.8 m around Rajrajeshwar, Hatiya, and Manipura by 2050, suggesting enhanced tidal energy and flow velocities that are likely to accelerate shoreline erosion and land loss, particularly along their southern and eastern margins. These findings highlight the increasing vulnerability of the Meghna Estuary's islands to climate change-driven hydrodynamic shifts, emphasizing the urgent need for targeted adaptive management, improved flood risk mitigation, and resilience-building measures to protect the region's communities and ecosystems from future inundation and erosion risks.